PROJECT PROPOSAL

Intern: Abdullah Aslam | Track: Software Development | Organization: Verstech Labs
Project: Student Records Manager | Week 1: 11-18 October 2026

1. Project Name and Purpose

Student Records Manager is a web-based application for schools and academies to organize student information and academic records in one place. It aims to replace repetitive manual record keeping with a searchable system for managing student details, marks and basic performance summaries.

2. Problem Statement

Paper registers and disconnected spreadsheets can make it difficult to find and update student information, avoid duplicates and calculate academic results consistently. The proposed application addresses these problems using structured digital records and controlled access.

3. Target Users

- Administrator: manages student records, oversees the system and controls authorized access.
- Teacher: searches student records, manages subject-wise marks and reviews performance within assigned permissions.

4. Main Features (7)

Feature	Description
1. Student records	Create, view, update and delete student details.
2. Search	Find students by name or roll number.
3. Marks management	Record and update subject-wise marks.
4. Automatic results	Calculate total marks, percentage and grade.
5. Authentication	Secure login and logout for staff.
6. Role-based access	Different permissions for administrator and teacher.
7. Dashboard	Display student totals and basic academic summaries.

5. Proposed Technology Stack

Component	Technology
Frontend	React.js (JavaScript)
Backend	Node.js and Express.js
Database	MongoDB with Mongoose
API testing	Postman
Version control	Git and GitHub

6. Project Scope and Outcome

The six-week implementation targets a small, working system covering the seven features above, supported by a documented database design, tests, a GitHub repository, and a demonstration. Attendance, fee payments, parent/student portals and messaging integrations are out of scope for this initial version.

7. Delivery Approach

The application will be developed incrementally: Week 1 - proposal, requirements and repository; Week 2 - database and structure; Week 3 - CRUD and core logic; Week 4 - authentication and additional features; Week 5 - testing and debugging; Week 6 - deployment (where applicable), documentation and final presentation.

Document status: Week 1 proposal prepared for review and submission. Project-specific design details may be refined during implementation.